

FINAL EXPLANATION: MATHEMATICS GRADE 10

Student Assessment Document

FINAL EXPLANATION: MATHEMATICS GRADE 10 – SUB-STRAND 1.1 REAL NUMBERS: FINAL EXPLANATION

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

You have finished the unit on real numbers. Now write your own Final Explanation to answer the Driving Question: Why can some numbers be written as exact fractions and others cannot, and how does knowing a number's type help us solve everyday problems in Kenya? Write your answer in the blank space under each prompt. Use evidence from your lessons: the Nairobi pizza diagrams, your fraction-circle results, your Number World Map, your reciprocal comparison table and your Driving Question Board. Include worked numbers where the prompt asks for them. Use full sentences and show your reasoning.

Part 1: The Pizza Puzzle and Whole-Number Types

Think back to the friends in Nairobi sharing a pizza. (a) Using your fraction-circle results, describe what happened to the decimal for one slice when the pizza was cut into 4 slices and when it was cut into 3 slices. (b) Whole numbers can be sorted as odd, even, prime or composite. Choose any three numbers from the set 2, 3, 4, 9, 15, 17, 36, 41, 49, 64 and say how you classified each and why. (c) Explain why a pizza cut into a prime number of slices, such as 3 or 7, is harder to share equally between 2 people than one cut into 4 or 8 slices. Mention what is special about the number 1 and the number 2.

Write your answer here:

Part 2: Rational and Irrational Numbers

(a) Explain what makes a number rational and what makes a number irrational. Talk about p/q and about what the decimal does. (b) Classify each of these as rational or irrational and give your reason: 0.75 , $1/3$, $\sqrt{2}$, π . (c) Some people use $22/7$ for π . Is $22/7$ equal to π ? Explain. (d) Use your number-line strip: between which whole numbers did $\sqrt{2}$ and π sit, and what does this tell you about a number that has no exact fraction?

Write your answer here:

Part 3: Reciprocals

(a) In your own words, what does the reciprocal of a number tell us in the pizza situation? (b) Find the reciprocal of $\frac{1}{4}$, of $\frac{2}{3}$ and of $\sqrt{2}$. Say whether each reciprocal is rational or irrational. (c) Does every real number have a reciprocal? Explain. (d) Amina in a mandazi problem has 1 kg of flour and uses $\frac{2}{3}$ kg per batch. Show how reciprocals help you find the number of batches.

Write your answer here:

Part 4: Choosing a Method and Predicting by Type

(a) Name the three methods you used to find reciprocals. (b) You found the reciprocal of $1 \div 7$ and of $1 \div \sqrt{2}$ using all three methods. What did you notice about the results for each, and why? (c) $\sqrt{4}$ is written with a radical sign. Is it rational or irrational, and what does this mean for its reciprocal? (d) Explain how knowing a number's type lets you predict something about its reciprocal before you calculate.

Write your answer here:

Part 5: Final Explanation and Kenyan Transfer

Answer the Driving Question in a connected explanation of 6 to 8 sentences: Why can some numbers be written as exact fractions and others cannot, and how does knowing a number's type help us solve everyday problems in Kenya? Use the Nairobi pizza, at least one Kenyan context from your lessons (for example mandazi in Kisumu, fertiliser bags in Eldoret, tiles in Mombasa, sharing land in the Rift Valley or githeri portions) and at least two key terms from this unit.

Write your answer here:

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Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Classification of numbers (whole, rational, irrational)	Correctly classifies all numbers, gives accurate definitions with p/q and decimal behaviour, and corrects the $22/7$ and $\sqrt{4}$ ideas.	Classifies most numbers correctly with mostly accurate definitions; may miss one subtle case such as $22/7$ or $\sqrt{4}$.	Confuses types or gives incomplete definitions with few reasons.
Reciprocals and calculation	Finds all reciprocals correctly, explains what a reciprocal means, notes zero has none, and solves the mandazi problem with clear steps.	Finds most reciprocals correctly and explains the meaning, with small gaps in steps or in the zero case.	Makes errors in reciprocals or cannot explain what they mean.
Use of evidence from the unit	Uses specific evidence such as fraction-circle decimals, number-line positions and method comparisons to support claims.	Uses some evidence from the unit but it is general in places.	Little or no reference to evidence from lessons.
Connecting number type to prediction and the phenomenon	Clearly explains how type predicts whether a reciprocal or slice is exact or approximate and links it to the pizza and Kenyan contexts.	Links type to exact or approximate answers, with a partial connection to the pizza or a Kenyan context.	Does not connect number type to prediction or to the phenomenon.
Mathematical communication	Writes clear, well-organised sentences using correct terms such as rational,	Writing is mostly clear with generally correct terms.	Writing is unclear or uses terms incorrectly.

	irrational, terminating, repeating and reciprocal.		
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ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

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